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# Are you in the room? A framework for sortition and representation

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# Copyright notice

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# Abstract

Sortition, the random selection of individuals to form deliberative panels, has seen a renaissance in recent years, as institutions realise the value of direct community and stakeholder input in deliberation and policymaking. These panels improve the quality of outcomes, increase public trust and engagement, and create more inclusive processes. Yet little attention has been paid to whether panels preserve a community's intersectional structure – how demographic traits combine – rather than only its marginal proportions. This thesis uses a simulated sortition pipeline (population, survey sample, and panel selection via the LexiMin algorithm) to test whether quota-constrained selection preserves the joint two-variable association present in a survey sample. Across all simulated conditions, effectively every feasible panel showed a two-variable relationship significantly different from its sample, at a rate far exceeding simple random sampling. These findings motivated Intersection Inspector, a diagnostic Shiny app helping practitioners assess a panel's representativeness before deliberation begins.

**Keywords:** sortition; citizens' assemblies; representativeness; intersectionality; panel selection algorithms

# Declaration

This thesis is an original work of my research and contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

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### Reproducibility statement

This thesis is written using Quarto with renv (Ushey & Wickham 2025) to create a reproducible environment. All materials (including the data sets and source files) required to reproduce this document can be found at the Github repository `github.com/Pian0Dan/Honours`. Analysis and simulation were carried out in R (R Core Team 2026), making use of the ggplot2 (Wickham et al. 2026a), ggmosaic2 (Friendly et al. 2026), patchwork (Pedersen 2025), dplyr (Wickham et al. 2026b), tidyr (Wickham, Vaughan & Girlich 2025), tidyverse (Wickham 2023), knitr (Xie 2025), kableExtra (Zhu 2026), and shiny (Chang et al. 2026) packages.

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The Intersection Inspector app code (`scripts/intersection_inspector/`) is licensed separately under the GNU Affero General Public License v3.0, which requires anyone who modifies and runs the app as a network service to make their modified source available to its users.

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I also believe thanks are in order for Bailey Flanigan and her team. Without your work, I would not have ended up on a community panel, getting the job I did, or where I am today as a whole, and would have had nothing to write about.

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I acknowledge the use of Posit Assistant (using an API form of Claude-sonnet) to assist me in the preparation of code, LaTex renders and initial drafting of section skeletons. The output from these was used to support my workflow in producing visualisations and clearly-written explanations.

# Chapter 1

# Introduction

## 1.1 Background

In a social landscape where the public is growing progressively disillusioned with its representation in politics (Landemore 2026), it is important to consider ways to foster more inclusive decision-making practices. One way of achieving this is using sortition to create panels that are more representative of the population for political discussions and deliberations. Adopted recently by an increasing number of institutions, including the Parisian ("Paris Creates a Permanent Citizens' Council" 2021), Irish (Walsh & Elkink 2021), and British Columbian (Ward 2008) governments, sortition as a part of democratic process is gaining traction. As a result, the need for fair and representative methods of building panels is only growing, and progress is being made in improving the structure of selective algorithms. However, there are still many challenges to overcome in order to ensure that these panels consistently achieve the level of representation that they envision. And there remains little procedure on measuring representativeness beyond the "surface" marginal level of demographics.

Sortition is a method of selection that uses random sampling to create panels that should be descriptively representative of a population. It has been used in various contexts, reaching as far back as ancient Athens (where it was how jurors were selected), and now into modern times for citizens' assemblies. The idea is that randomly selecting individuals from the population can create panels that are more representative and inclusive than those selected through traditional methods, like elections or open-ended surveys. Due to its random nature, sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), making it more likely that decisions are in the relevant population's best interest.

### 1.1.1 The representation problem

One of the main roadblocks in sortition is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of what it means to be fair, what it means to be representative, and how these look from a systematic perspective. A fair lottery is not, by itself, proof of a representative outcome – it answers whether everyone had a reasonable chance of being chosen, not whether the people who actually get chosen reflect the community they are meant to stand in for. This thesis is concerned less with whether the process was fair, and more with a plainer question underneath it: who actually ends up on a panel?

As a process, sortition moves through several distinct stages between defining a population and convening a panel (Chapter 2 breaks these down in full), and each one has the potential to introduce bias that compounds by the end – yet there is currently no rigorous evaluation of representation achieved before a deliberative process is run. It is therefore crucial to measure how the level of representativeness (or *potential* representativeness) flows through each stage, and to understand how different factors may affect it.

### 1.1.2 Progression of sortition methods

Sortition methods have evolved substantially over time, moving from simple self-selection and proportional sampling toward increasingly systematic algorithmic approaches – LEGACY, MaxiMin, MiniMax, and most recently LexiMin and Goldilocks – each refining how fairly and robustly individual selection probabilities are equalised across a respondent pool (Chapter 2 gives the full history and formal definitions). Of these, LexiMin is the algorithm in practical use today, running inside Panelot (Flanigan et al. 2020), the tool currently used to assemble real sortition panels, and is the algorithm this thesis works with directly.

## 1.2 Thesis scope

All of the aforementioned algorithms only focus on optimising for maximal selection likelihoods as their sole objective functions. This means that in its discovery and construction of feasible panels, the existing LexiMin algorithm may be prone to distortion of intersections and intersectional association trends present within population distributions. This presents a significant risk for the validity of representation achieved by panels output by the algorithm – the extent and reasoning of which will be explored further in the paper.

Additionally, there is limited literature on the visualisations of the outputs of these algorithms. This

presents a great opportunity to demystify LexiMin and sortition as a process, improving transparency for practitioners, facilitators, participants, and broader community stakeholders alike. The comparative visualisation methods and tools also allow a general comparison to be made between any selected group and any broader sample it is pulled from, with any algorithm or selection process. This could be useful as a mechanism for quick sanity checks of panel configurations, and as a way for practitioners to evaluate the validity of panels before engaging in deliberation.

To investigate this empirically, this thesis builds a simulated sortition pipeline: synthetic populations with a known, deliberately introduced two-variable association (and a matched population with none), survey samples drawn under both equal-chance and differential response, and panels selected from those samples via LexiMin. Statistical tests then compare each panel's joint two-variable structure against its own sample, isolating whether a panel's intersectional composition diverges from the sample it was drawn from beyond what sampling variability alone would produce. The resulting findings motivate a diagnostic Shiny app, Intersection Inspector, that lets practitioners check this directly for their own panels.

### 1.2.1 Thesis structure

The remainder of this thesis is structured as follows: Chapter 2 provides a literature review of the existing research on sortition and representation, and identifies gaps in the current knowledge. Chapter 3 outlines the simulation-based methodology used to evaluate panel representativeness, including how differential survey non-response is modelled alongside an equal-chance baseline, and the statistical tests used to assess whether a panel's joint category distribution preserves the association present in the sample it was drawn from. Chapter 4 presents the results of the analysis, showing near-total intersectional association distortion in selected panels, well beyond what a random-sampling benchmark would produce. Chapter 5 introduces "Intersection Inspector", a tool developed in response to those results to help practitioners assess panel representativeness directly. Chapter 6 concludes by proposing a framework and recommendations for practitioners, discussing the limitations of this thesis, suggesting directions for future research, and reflecting on what genuine representation demands beyond marginal fairness – namely, keeping sight of who the process is meant to represent, and whether it actually does.

# Chapter 2

# Literature review

## 2.1 Representation and sortition

### 2.1.1 What does it mean to be representative?

Every sortition selection method shares the same basic aim: a panel that is fair and representative of the population. But what does that actually mean?

Fairness means the selection process is transparent, unbiased and inclusive – everyone should have an equal chance of being chosen, regardless of their background or demography. In practice, this comes down to minimising the gap in selection likelihood across the pool, typically achieved by introducing randomness into the process (Flanigan et al. 2021); without it, selection bias creeps in implicitly.

Representativeness, meanwhile, is about whether the selected panel actually reflects the demographics and perspectives of the population it was drawn from – diverse, inclusive, and populated by people from the range of backgrounds the process set out to capture. This can be checked directly, by comparing a panel's demographic makeup against the population's.

Approaching it procedurally, the sortition process can be thought of in four stages:

- Defining the population and traits of interest
- Sampling from a random subset of the population (through survey or other outreach)
- Panel selection (through algorithmic means or other method)
- Panel evaluation, confirmation and engagement

Each step is a chance for bias to creep in and representativeness to suffer, which is why it matters to evaluate representativeness (or potential representativeness) at every stage, not just the end.

If the population and the traits of interest are poorly defined, for instance, the selection process may fail to capture the group's actual diversity, leaving the panel less representative. This is a real challenge in itself: most population descriptions, as discussed above, record demographic characteristics separately rather than capturing how they intersect, which is central to representativeness. Addressing that gap properly would mean wading into broader data collection and storage practices, which is beyond this thesis's scope – but it is worth keeping in mind that representation goes beyond the margins.

The same applies to survey response: if certain groups are less likely to respond, they end up underrepresented in the pool of potential panelists before a panel is even assembled. If a survey of an area that is 60% women somehow drew no female respondents at all, representativeness would already be violated well before selection begins. Cases like this raise the question of how to account for non-response bias, and whether visualisation can help show the gap between the respondent pool and the population – which is part of what this thesis sets out to do.

### 2.1.2 Panel selection

Suppose there is a pool of individuals numbered 1 to $n$, from which a panel of size $k$ is to be chosen. Let $K$ be the set of all feasible panels – any group of $k$ individuals satisfying the required constraints, which generally take the form of quotas.

Quotas specify a desired count, or range of counts, for individuals in a given trait class (e.g. 20-22 males, or 15-17 rural), tied to whichever demographics a project has deemed important to represent. The core goal of sortition, then, is to build a panel that meets those quota constraints equitably, robustly, and in a way that genuinely represents the population it was drawn from.

### 2.1.3 Existing methods

#### Non-algorithmic/self-selection

This is the simplest method of panel selection: individuals are simply allowed to self-select, for example by advertising the opportunity and taking sign-ups on a first-come, first-served basis. These methods tend not to produce representative panels, for two reasons. First, they invite self-selection bias – involvement depends on exposure rather than random chance, and the people who put their hand up are more likely to already hold formed opinions, ahead of ever joining the panel. Second, they tend toward lower engagement, since there is no guarantee of reaching a particular level of exposure in the first place.

#### LEGACY heuristic (Flanigan et al. 2021; Sortition Foundation 2024)

Already an improvement on non-algorithmic methods, the LEGACY selection heuristic builds a panel by selecting respondents one at a time until it is full. At each step, it calculates a "need" value to find

the trait currently furthest from its desired count, then picks a random individual who has that trait. If selecting someone would push a trait past its upper quota, everyone else with that trait is removed from the pool before the next pick. This repeats until the panel is complete, or the process runs out of eligible individuals and restarts. LEGACY satisfies the basic requirement of meeting quotas, but it does not maximise representativeness or fairness: it ignores each individual's actual selection likelihood (which, in extreme cases, can end up at 0%) and makes no attempt to minimise the maximum deviation from an equal chance of selection. For a time, this was the Sortition Foundation's own method of panel selection, implemented in their GroupSelect tool, before the algorithmic fairness methods below were developed and adopted.

#### MaxiMin (Flanigan, Kehne & Procaccia 2021) and MiniMax (Baharav & Flanigan 2024)

This shortcoming is addressed by the first two algorithmic sortition methods, MaxiMin and MiniMax, which tackle the problem in opposite ways. MaxiMin generates potential panels and selection probabilities, then sets out to maximise the *minimum* selection probability across the pool. This reduces the chance of any individual never being picked, improving fairness, but makes the method vulnerable to manipulation – someone who misreports belonging to an under-represented quota group can inflate their own odds. MiniMax instead minimises the *maximum* selection probability, making selection more robust against manipulation, but at the cost of fairness, since the risk of a 0% selection chance returns.

#### LexiMin (Flanigan et al. 2021)

LexiMin – short for Lexicographic MaxiMin – works similarly to MaxiMin, but resolves ties sequentially rather than stopping at the single minimum. It first identifies whoever has the lowest selection chance and maximises that, then moves to the next-lowest, and so on, until every individual's selection probability has been addressed and made as equal as possible across participants. This gives everyone the highest chance of selection achievable without disadvantaging someone already prioritised – but, like MaxiMin, it inherits a vulnerability to manipulation.

#### Goldilocks (Baharav & Flanigan 2024)

This brings the discussion to the imminent gold standard: the aptly-named Goldilocks algorithm. Goldilocks simultaneously maximises minimum selection probabilities (for fairness) and minimises maximum selection probabilities (for robustness), aiming for a "best of both worlds" balance between MaxiMin and MiniMax. The motivation is clear: both MaxiMin and LexiMin place no ceiling on how high an individual's selection probability can climb, so a participant who misreports their trait membership to appear under-represented can disproportionately inflate their own odds. By capping the maximum selection probability, as MiniMax does, Goldilocks limits the payoff from that kind of manipulation while still prioritising the lowest probabilities enough to preserve most of MaxiMin

and LexiMin's fairness guarantees – a selection rule that is simultaneously hard to manipulate and reasonably fair.

### 2.1.4 Why focus on LexiMin

Goldilocks may be the newer and, in theory, the better algorithm, but this thesis works with LexiMin instead. The reasons are practical. LexiMin is the algorithm actually in use: it runs inside Panelot (Flanigan et al. 2020), the tool practitioners currently use to put real panels together, while no public version of Goldilocks existed at the time of writing. Testing LexiMin means testing the algorithm actually shaping panels today, not one that only exists on paper so far.

It also makes little difference to the specific problem this thesis is looking at. Goldilocks was built to fix manipulation, not to fix how panels handle intersections of traits – and on that front, it works the same way LexiMin does. Both algorithms set selection probabilities purely from marginal quotas, and neither one looks at how traits combine. Capping the highest selection probability, which is all Goldilocks adds on top of LexiMin, has no obvious reason to change how well a panel's mix of traits reflects its sample. So the gap this chapter identifies below should apply to Goldilocks just as much as it applies to LexiMin. Working with LexiMin here is a matter of what could actually be tested and what practitioners actually use, not a claim that it matters more – the same argument should carry over to Goldilocks once it can be tested directly.

#### LexiMin approach (Flanigan et al. 2021)

Rather than committing to a single panel immediately, LexiMin constructs a probability distribution over the full set of feasible panels $K$, assigning each panel $S \in K$ a non-negative probability $p_S$ that sums to one across the set. An individual's overall selection chance is then the sum of the probabilities of every panel that includes them. Sorting these chances from smallest to largest, LexiMin proceeds lexicographically: it maximises the smallest probability first, then – holding that value fixed – maximises the second-smallest, and so on until every individual has been addressed.

### 2.1.5 Formal representation, algorithm and logic

This process can be expressed formally. Let $p_S$ denote the probability assigned to panel $S$, which must be non-negative and sum to one across every feasible panel in $K$,

$$p_S \geq 0, \qquad \sum_{S \in K} p_S = 1. \tag{2.1}$$

The resulting selection probability for an individual $i$ is the sum of the probabilities of every feasible panel containing them, giving the full vector of individual selection probabilities across the population,

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S, \qquad \pi(p) = (\pi_1(p), \pi_2(p), \ldots, \pi_n(p)). \tag{2.2}$$

Sorted from smallest to largest, $\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq \ldots \leq \pi_{(n)}(p)$, LexiMin chooses the distribution $p$ that maximises this vector in lexicographic order – the smallest entry first, then the second-smallest subject to the first remaining maximised, and so on down the vector.

In practice, this runs as a sequence of steps: every feasible panel in $K$ is listed alongside every possible probability distribution over those panels, and each candidate distribution's resulting selection probabilities are computed and sorted as above. Only the distribution (or distributions) that maximise the smallest probability $\pi_{(1)}(p)$ are retained; if several remain tied, this narrows further to those that also maximise the second-smallest, and so on, resolving ties one probability at a time until a single distribution remains, from which a final panel is then selected.

At its core, then, LexiMin is purely a fairness rule: it prioritises whoever has the lowest chance of selection, improves their odds first, then works through the next-lowest probabilities until no further improvement is possible without disadvantaging someone already prioritised – leaving selection probabilities as equal as possible across the population, within the panel's quota constraints.

## 2.2 Visualisation methods

Current sortition practice communicates a sample or panel's composition almost exclusively through pie charts, or bar charts showing the same information in a different shape – one chart per demographic category, each showing that variable's marginal breakdown in isolation. Both are easy to produce and easy for a non-specialist audience to read at a glance, which likely explains their popularity with practitioners reporting on panel composition. But neither can encode more than a single categorical variable at a time: there is no visual channel left over to show how two variables relate to one another. For example, a practitioner working only from pie or bar charts can confirm that a panel's age breakdown looks right, and separately that its tenure breakdown looks right, but has no way of seeing whether the right *combinations* of age and tenure made it onto the panel. Faceting a bar chart by a second variable does not fix this: each facet's bars are rescaled relative to that facet alone, rather than the whole population, so an association between the two variables – which can only be seen when proportions stay relative to the full population – is exactly what gets obscured by splitting the chart apart in the first place.

Mosaic plots are a longstanding answer to exactly this limitation. A mosaic plot divides a rectangle into tiles whose widths and heights are proportional to the categories of two (or more) variables, so

that each tile's area represents the joint frequency of that specific combination, rather than either variable's marginal frequency alone (Hofmann 2003). Graphical methods for categorical data are comparatively underdeveloped next to those available for numeric data, and mosaic plots have been established in the statistical graphics literature as one of the more effective options for visualising relationships between multiple categorical variables at once. Because tile area scales with both variables simultaneously, a departure from proportional tiling signals that the two variables are not independent, in a way no pie chart or ordinary bar chart can show. The current R package for building mosaic plots within the ggplot2 grammar of graphics is ggmosaic2 (Friendly et al. 2026), an actively maintained extension of the original ggmosaic implementation (Jeppson, Hofmann & Cook 2026).

## 2.3 Discussion and new investigation

This approach is effective at making selection probabilities as equal as possible, which helps make sortition fairer by reducing situations where specific individuals are disadvantaged. It does not, however, provide any way to impose joint trait requirements on subsets of the population – something that may be necessary for accurate representation of community groups. That gap reflects a broader pattern across this literature: algorithmic effort has gone almost entirely into optimising the fairness and robustness of the selection *mechanism*, rather than validating the *outcome* that mechanism produces. A panel satisfying every fairness and robustness guarantee an algorithm offers is not thereby guaranteed to represent the population it was drawn from well – these are related but distinct properties, and the literature so far has focused heavily on the former at the expense of the latter, undermining representativeness as a goal given the motivation for sortitioned panels in the first place: filling a room with individuals who accurately and descriptively reflect a community's population diversity and characteristics.

One specific way this gap shows up: no algorithmic technique surveyed here stratifies for combinations of demographic traits in its decision-making – each instead affects individual selection chances purely through quotas set on each trait at the margin. This is a real limitation, since it means the selection process may fail to capture the group's actual diversity, leaving a less representative panel. In some situations – side-panels built for specific subgroups, or communities with known intersectional minorities of interest – ensuring specific intersectional combinations are included may be necessary for genuine representation. This gap is generally patched in practice by bringing in "specialist speakers" during deliberation (Redman & Carson n.d.), but that is a different kind of inclusion than being part of the panel and its decision-making directly.

Notably, this gap is rarely argued for directly; it is largely left unaddressed. The implicit assumption

running through this literature appears to be that satisfying marginal quotas is enough – that selecting individuals to hit the right counts for each trait independently will, through the randomness built into the selection mechanism, naturally produce a panel whose *combinations* of traits are also reasonably representative, without this needing to be checked or enforced directly. None of the algorithmic methods reviewed above test whether this actually holds, and the literature more broadly does not appear to examine it either.

This thesis treats that assumption as an open empirical question rather than a safe default. The simulations that follow test how much intersectional inclusion happens naturally under current methods, and establish a way to measure how much intersectionality and association survives the selection process.

# Chapter 3

# Methodology

## 3.1 Research aims

This thesis set out to assess whether current algorithmic methods achieve representative panels, diagnose whether a sample's association trends survive the sortition process, demystify sortition selection to improve its transparency, and provide a framework for more informed, standardised practitioner decisions.

Underlying these aims is a more basic question: what does it mean for a panel to be "good" at representation, and are current methods actually achieving it? That breaks down further. Are algorithms like LexiMin performing as well as they could? How well do intersectional associations and trends survive selection, and therefore how representative is panel composition? And how much can differing survey response rates and other sampling-level issues bias representativeness, beyond whatever the algorithm itself contributes?

## 3.2 Research design

### 3.2.1 Simulation design

To test these questions, two populations of simple intersectional complexity (two variables) with varying associations (independent and associated) were simulated. The findings should generalise to cases with more categories, and are expected to be more severe there, since the intersectional matrix only grows in dimension while panel size stays fixed. The categories used – age (18-34, 35-44, 45-54, 55-64, 65+) and tenure status (rent, own, other) – were chosen to mirror demographic variables commonly used for sortition in practice. Combining these two variables produced the four simulated sample/respondent datasets used for panel selection.

### 3.2.2 Population, slice, sample and panel: procedure and definitions

Every simulated dataset in this thesis moves through the same four stages, with terms kept deliberately consistent throughout to avoid ambiguity – "sample" in particular is used loosely, and inconsistently, across much of the sortition literature.

- **Population**: the full simulated universe of individuals ($n_{\text{total}} = 100{,}000$), generated from a specified joint probability array over the category variables. This represents the entire community eligible for selection, with marginal and joint (intersectional) distributions known exactly, since they were specified directly in the data-generating process.
- **Slice**: a simple random sample, without replacement, of 10% of the population, drawn before any response behaviour is simulated – representing the group actually invited to take part (e.g. mailed a survey), sometimes called the *invited pool* elsewhere. Because it is drawn uniformly at random, the slice's composition deviates from the population's only through ordinary sampling variability, not systematic bias.
- **Sample**: the subset of the slice who actually respond, determined by an independent Bernoulli trial for each invited individual under one of the response-rate cases described below. This is the *realised* pool available for panel selection – also called the *respondent pool* or *survey sample* elsewhere – and the benchmark every panel's representativeness is compared against, since it, not the full population, is what any practical algorithm actually draws from.
- **Panel**: a fixed-size subset of $k$ members drawn from the sample by a selection algorithm (LexiMin, via Panelot, in current practice) subject to quota constraints on each category level's permitted count – the final body that would convene to deliberate, and the unit whose representativeness this thesis is ultimately concerned with.

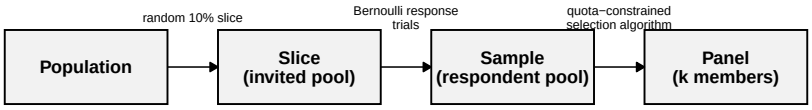



**Figure 3.1:** *The four stages of the simulation pipeline: population, slice, sample and panel.*

Figure 3.1 summarises this pipeline. Each stage can introduce its own departure from the population's true composition: the slice through ordinary sampling variability, the sample through differential response rates, and the panel through the selection algorithm's own quota mechanism (Section 3.2.5). Distinguishing these stages explicitly makes it possible to attribute any observed distortion in a final panel to a specific stage of the pipeline, rather than to sortition "in general".

### 3.2.3 Population construction: age and tenure

To illustrate the general procedure concretely at a minimal scale, this thesis builds a simple two-variable example using only age and tenure. Age is denoted $A$, taking one of $L_A = 5$ ordered levels $a \in \{1, \ldots, 5\}$ (18-34 through 65+), and tenure is denoted $T$, taking one of $L_T = 3$ levels $t \in \{1, \ldots, 3\}$ (Rent, Own, Other). Marginal probability vectors were defined for each variable,

$$\mathbf{p}_A = \left(p_{A_1}, \ldots, p_{A_5}\right), \qquad \mathbf{p}_T = \left(p_{T_1}, p_{T_2}, p_{T_3}\right), \qquad \sum_{a=1}^{5} p_{A_a} = 1, \qquad \sum_{t=1}^{3} p_{T_t} = 1, \tag{3.1}$$

with $\mathbf{p}_A = (0.15, 0.20, 0.25, 0.25, 0.15)$, a mild skew toward the upper-middle age brackets, and $\mathbf{p}_T = (0.30, 0.50, 0.20)$, a moderate skew toward home ownership. Both the independent and associated populations below use these same marginal targets, so any difference between them is attributable solely to the association introduced, not to a difference in either variable's marginal distribution.

An independent population (`sim_data2`) was generated by taking the outer product of these two marginals to form a joint probability array,

$$P^{(0)}(a, t) \;=\; p_{A_a}\, p_{T_t}, \qquad \mathbf{P}^{(0)} \;=\; \mathbf{p}_A \mathbf{p}_T^{\top} \;\in\; \mathbb{R}^{5\times 3}, \tag{3.2}$$

a $5 \times 3$ matrix whose row sums equal $\mathbf{p}_A$ and column sums equal $\mathbf{p}_T$ by construction, so that $A \perp T$. This array was flattened into a probability vector and cell counts were drawn from a single multinomial trial before being expanded into individual-level records, as described in full below.

A second, associated population (`sim_data2_cov`) uses the same marginal targets but deliberately introduces a negative association between age and renting, via an ordinal score for age,

$$s(a) = \frac{a-1}{5-1} \in [0, 1], \tag{3.3}$$

running from 0 for the youngest bracket to 1 for the oldest, which was used to seed the independent array $\mathbf{P}^{(0)}$ with an exponential adjustment factor targeting only the "Rent" level of tenure,

$$\tilde{P}(a, t) \;=\; P^{(0)}(a, t) \cdot \exp\Big(-\alpha\, s(a) \cdot \mathbb{1}[t = \text{Rent}]\Big), \qquad \alpha = 1.2, \tag{3.4}$$

scaling down the probability of renting by a factor of $e^{-\alpha s(a)}$ that grows with age, while leaving owning and other tenure unchanged. Because this adjustment only touches the "Rent" column, the row and

column sums of $\tilde{\mathbf{P}}$ no longer equal $\mathbf{p}_A$ and $\mathbf{p}_T$, so the array was rescaled through iterative proportional fitting (raking) to restore the age and tenure marginals exactly to their specified targets. Repeating this two-pass rescaling for $R = 100$ iterations (well past convergence for a $5 \times 3$ table; full update rule in Appendix A) yields a raked array $P^\star$ satisfying $\sum_t P^\star(a,t) = p_{A_a}$ for all $a$ and $\sum_a P^\star(a,t) = p_{T_t}$ for all $t$, to numerical tolerance, while retaining the negative age-renting association introduced by the exponential seeding step – raking restores the specified marginals while preserving the association the adjustment introduced.

Both populations were sampled the same way, differing only in which array was used. The relevant array ($\mathbf{P}^{(0)}$ for the independent case, or the raked $P^\star$ for the associated case) was flattened into a probability vector $\mathbf{p}^\star \in \mathbb{R}^{15}$ (age varying fastest), and cell counts were drawn from a single multinomial trial,

$$(n_1, \ldots, n_{15}) \sim \text{Multinomial}(n_{\text{total}}, \mathbf{p}^\star), \qquad n_{\text{total}} = 100{,}000, \tag{3.5}$$

with each cell $j$, corresponding to one $(a,t)$ combination, expanded into $n_j$ individual-level records sharing that age and tenure. Comparing the two resulting populations demonstrates the core logic of the simulation framework in its simplest form: a chi-squared test of association between age and tenure is non-significant in the independent population but highly significant in the associated population, with the conditional probability of renting falling steadily from around 44% among 18-34 year olds to around 18% among those aged 65 and over, even though both populations share identical marginal distributions for age and tenure individually. This confirms the seeding-and-raking procedure successfully introduces a targeted association without disturbing the overall marginal proportions, and sets up a direct test of whether the selection algorithm preserves intersection structure beyond the purely marginal level.

### 3.2.4 Slice and sample construction: response simulation

As is standard practice, a random slice of the population (10%) is taken to serve as the survey sample, from which responses are drawn. Concretely, from a population of $n_{\text{total}}$ individuals, a survey sample $\mathscr{S}$ of size $n_{\mathscr{S}} = \lfloor 0.10\, n_{\text{total}} \rceil$ (so $n_{\mathscr{S}} = 10{,}000$ for the two-variable example, drawn uniformly at random without replacement) is taken as the invited pool, before any response behaviour is simulated. Whether each invited individual $i \in \mathscr{S}$ actually responds is then modelled by a binary response variable $R$, with $R_i = 1$ if individual $i$ responds and $R_i = 0$ otherwise, drawn as an independent Bernoulli trial,

$$R_i \overset{\text{ind.}}{\sim} \text{Bernoulli}(\pi_i), \tag{3.6}$$

where the response probability $\pi_i$ is assigned according to one of two cases: an independent case, and an age-covaried case mirroring the age-renting association introduced at the population level above.

In the *independent* case, every invited individual is equally likely to respond regardless of their trait profile,

$$\pi_i = \pi_0, \qquad \pi_0 = 0.30, \tag{3.7}$$

a baseline against which the effects of differential response can be compared; a chi-squared test confirms no significant association between $A$ and $R$ here, as expected. In the *age-covaried* case, response likelihood increases with age, reusing the same ordinal score $s(a)$ defined above,

$$\pi_i = \pi_1 \exp\big(\alpha_R s(a_i)\big), \qquad \pi_1 = 0.20, \quad \alpha_R = 1.0, \tag{3.8}$$

so that the response rate rises from around 20% among the youngest bracket to over 50% among the oldest, and the association between $A$ and $R$ is now highly significant.

Applying both response cases to both populations produces four respondent-pool files – `sim_data2_ind_eq.csv`, `sim_data2_ind_ass.csv`, `sim_data2_cov_eq.csv` and `sim_data2_cov_ass.csv` – each containing only the individuals who actually responded under that population/response-case combination, with their age and tenure. These names follow the same `<ind|cov>_<eq|ass>` convention used for the lottery files below, so a file name alone identifies both the population and response case it came from. Each respondent pool is, in the terminology above, a *sample*: the realised pool a panel can actually be drawn from, and the input to the panel-selection step described next.

### 3.2.5 Panel generation and evaluation

#### Panel generation and representation

With the four respondent-pool files constructed above, the final stage of the pipeline draws an actual panel from each respondent pool. This was done using Panelot, a sortition tool that selects panels of a fixed size $k$ (here $k = 50$) subject to quota constraints on each variable's margin independently – a minimum and maximum permitted count for every age bracket and tenure category, based on the population's marginal distributions and specified in a `categories.csv` file. For brackets whose exact

proportional target at $k = 50$ was non-integer – 18-34, 45-54, 55-64 and 65+, each targeting 7.5 or 12.5 members – the quota range was set to the floor and ceiling of that target (e.g. 7-8 for a 7.5 target). The remaining categories – 35-44 and all three tenure levels – land on exact integer targets and so carry a fixed quota with no slack. Rather than a single fixed panel, Panelot's output, `lottery.csv`, enumerates every feasible panel its selection algorithm discovered, together with the exact probability $q_j$ panel $j$ would be returned were the lottery run, so $\sum_j q_j = 1$ across the $J$ discovered panels. Each row lists the pool identifiers of the $k = 50$ selected respondents.

A deliberate feature of this quota structure is that it constrains the age and tenure margins of a panel separately, with no quota placed on their joint (intersectional) distribution. Two panels can therefore satisfy identical quotas – the same count of 18-34 year olds, the same count of renters – while differing substantially in which individuals combine which age with which tenure. This asymmetry between marginal and joint representativeness motivates the association-level evaluation developed below.

Four lottery files were generated in this way, covering every combination of the two populations (independent and associated) and the two response-rate cases (equal-chance and age-associated) introduced previously: `lottery_ind_eq.csv`, `lottery_ind_ass.csv`, `lottery_cov_eq.csv` and `lottery_cov_ass.csv`. Each panel's members are identified only by pool identifier in `lottery.csv`, so their age and tenure values were recovered by joining back onto the respondent-pool CSV originally supplied to Panelot (e.g. `sim_data2_cov_eq.csv`). This respondent pool – the people who were invited to respond and actually did so – is referred to throughout as the *survey sample*, and serves as the benchmark against which every panel's composition is compared.

### Panel-versus-sample homogeneity test

For each discovered panel $j$, a $5 \times 3$ contingency table of age against tenure was constructed from its $k = 50$ members, and a chi-squared test of independence applied to assess whether age and tenure appear related within that panel alone. With only 50 observations spread across 15 cells, several expected cell counts fall below the conventional rule-of-thumb minimum of five, so a Monte Carlo approximation to the null distribution (`simulate.p.value = TRUE`, $B = 2000$ resamples) replaced the standard asymptotic chi-squared reference distribution.

This within-panel test alone cannot establish whether a panel's association differs from the survey sample's – a panel could show a significant association purely because the sample itself does. To test this directly, each panel $j$ was compared against the remainder of the survey sample via a three-way log-linear model. Writing $G$ for a binary group indicator (panel member vs. not), a Poisson model was fitted to the cell counts of the $5 \times 3 \times 2$ Age $\times$ Tenure $\times$ $G$ table, first without the three-way interaction,

$$\log \mathbb{E}[\text{Freq}] = \beta_0 + \beta_A^{(a)} + \beta_T^{(t)} + \beta_G^{(g)} + \beta_{AT}^{(a,t)} + \beta_{AG}^{(a,g)} + \beta_{TG}^{(t,g)}, \tag{3.9}$$

and then with it,

$$\log \mathbb{E}[\text{Freq}] = \beta_0 + \beta_A^{(a)} + \beta_T^{(t)} + \beta_G^{(g)} + \beta_{AT}^{(a,t)} + \beta_{AG}^{(a,g)} + \beta_{TG}^{(t,g)} + \beta_{ATG}^{(a,t,g)}. \tag{3.10}$$

A likelihood-ratio test between these two nested models isolates the three-way interaction term $\beta_{ATG}$, with the test statistic $G^2 = 2(\ell_{\text{full}} - \ell_{\text{reduced}})$ referred to a $\chi^2_8$ distribution – 8 degrees of freedom corresponding to the $(5-1)(3-1)(2-1)$ free parameters contributed by $\beta_{ATG}^{(a,t,g)}$ across the $5 \times 3 \times 2$ table. Rather than testing whether the fitted $\hat{\beta}_{ATG}$ is itself close to zero, the test asks whether allowing the age-tenure association to differ between panel and non-panel members improves the model's fit by more than chance alone would produce. A significant result indicates that the age-tenure association differs between panel members and the rest of the sample, i.e. that panel $j$'s intersectional composition is not merely a smaller, noisier version of the sample's, but systematically different from it.

Because every panel discovered for a given lottery file is tested this way – several thousand panels per file – the homogeneity test was repeated many times over. Results are reported at an uncorrected threshold of $\alpha = 0.05$ throughout.

A panel of only 50 members is itself a small sample, so some disagreement with the full survey sample's association is expected from sampling variability alone, independent of anything specific to Panelot's mechanism. To separate a genuine quota-constrained-selection effect from this baseline noise, the same homogeneity test was applied to 500 simple random samples of size 50, drawn without quota constraints from the same survey sample. If panels were no more divergent from the sample than equivalently sized random draws, the proportion flagged as significantly different should resemble the proportion of random samples flagged – approximately $\alpha$ under the null.

Alongside each test, Cramer's V,

$$V = \sqrt{\frac{\chi^2/n}{\min(r-1,\, c-1)}}, \tag{3.11}$$

was computed from the within-panel contingency table, with $r = 5$ age brackets as rows and $c = 3$ tenure categories as columns (an arbitrary orientation choice, since $\min(r-1, c-1)$ is unaffected by

swapping which variable is assigned to rows versus columns), to express the strength, rather than merely the statistical significance, of the association detected within each panel.

### 3.2.6 Validating the simulation: numerical summary

Alongside the mosaic plots used throughout this thesis to compare the age x tenure population, slice and respondent sample visually (Chapter 2; visualisation methods are discussed further below), each simulated dataset is also checked numerically, to confirm the data-generating process behaved as intended before any downstream panel evaluation. Two quantities recur throughout this checking, the same chi-squared logic underlying the panel-level homogeneity test described above (Section 3.2.5):

- A **chi-squared test of independence**, applied to the $5 \times 3$ age-by-tenure contingency table, confirms whether the intended association (or lack of it) was actually realised in a given dataset. A non-significant result ($p > 0.05$) for the independent population (`sim_data2`), alongside a significant result for the associated population (`sim_data2_cov`), is the expected pattern – a sanity check on the simulation code, not a substantive finding in its own right.
- Marginal **proportion tables** (`prop.table()` applied to the age and tenure frequency tables) confirm that each variable's marginal distribution matches its specified target ($\mathbf{p}_A$ or $\mathbf{p}_T$) to within sampling error, at both the population and sample level. This matters particularly for the associated population, where raking is specifically designed to restore exact marginal targets despite the association-inducing adjustment applied beforehand.

### 3.2.7 Visualisation methods

The choice of visualisation follows from the nature of the data. The variables of interest – age brackets, tenure status, and other demographic categories used for sortition – are categorical rather than continuous, and the central question is how they intersect, not how any one is distributed alone. Scatterplots and histograms are built for continuous data; a bar or pie chart shows one categorical variable's marginal distribution but not how two or more co-occur (Section 4.1); and a bar chart grouped by a second variable becomes hard to read once cell counts are uneven – precisely this thesis's situation – because bar heights are not comparable the way proportional tile areas are. The mosaic plot, by contrast, represents both marginal and joint distributions within a single figure (Chapter 2), which is why it is the primary visualisation tool used throughout this thesis: it shows not just whether individual categories are represented in the right proportions, but whether particular *combinations* are over- or under-represented relative to population demographics. Mosaic plots were constructed using the ggmosaic2 package (Friendly et al. 2026), built within the ggplot2 grammar of graphics.

To make later plots easy to interpret, the simplest form of a mosaic plot is shown in Figure 3.2 below.

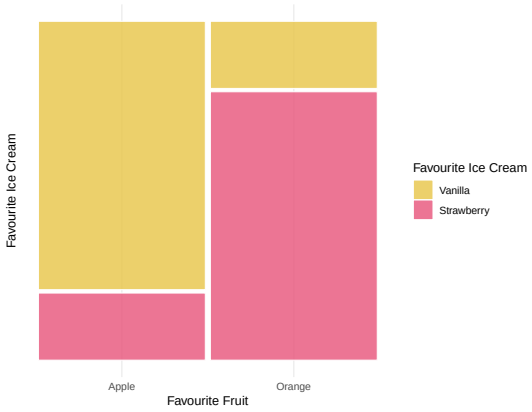



**Figure 3.2:** *A simple mosaic plot of favourite fruit (x-axis) against favourite ice cream flavour (fill), illustrating the mosaic plot convention used throughout this thesis. The two bars are split in opposite directions, illustrating a clear association between the variables.*

This example uses two variables, favourite fruit and favourite ice cream, with 25 individuals per fruit (50 total) and ice cream deliberately associated with fruit – 80% of Apple-lovers prefer Vanilla, 80% of Orange-lovers prefer Strawberry. Bar width reflects each category's marginal share (equal here); fill height reflects the conditional proportion of ice cream given fruit. The association appears in the fill split, which reverses almost entirely between bars, so fruit preference is highly informative about ice cream preference. Had the variables been independent, both bars would show close to a 50/50 split regardless of bar width: a *difference* in fill proportions, not bar width, is what signals an association, as with the independent and associated age-tenure populations introduced above.

This convention – x-axis width proportional to marginal share, fill height proportional to conditional proportion – is used throughout for the population, slice and sample comparisons, with age brackets on the x-axis and tenure as fill.

### 3.2.8 Method limitations and caveats

The response rates used for the age x tenure pipeline – 30% baseline, rising above 50% in the age-covaried case – are considerably higher than typical real-world sortition response rates, often below 10%. This was deliberate: lower rates, applied to a 10,000-person slice, would leave too few respondents in some of the 15 age-by-tenure cells to reliably assess quota satisfaction or intersectional preservation at $k = 50$, leaving the algorithm data-starved regardless of its own effectiveness. Elevating the response rate instead isolates the algorithm's own behaviour as the object of study, so any representativeness issues identified here are a conservative, best-case estimate: realistic response rates would leave even less data to work with, and more room for poorly representative panels.

Every association introduced in the pipeline – age-renting at the population level ($\alpha = 1.2$) and age-response at the sampling level ($\alpha_R = 1.0$) – used a single exponential strength parameter, fixed

at a moderate-to-strong value chosen to be clearly visible in a mosaic plot and reliably detectable by a chi-squared test. These values are not derived from any real-world dataset, and performance is unlikely to degrade uniformly with association strength: a weaker but evenly spread association might survive a process that fails here, while an association resting on fewer individuals could be disrupted more easily even if comparably strong. Because this simulation uses fixed strengths rather than a systematic sweep, the panel-level results in Section 4.3 should be read as showing that failures of this kind are possible and can be severe at the strength simulated, not as a generalisable estimate of their magnitude in practice.

A further question is what a panel's divergence from its sample actually shows, not just how large it is. Panelot's quotas come from each category's *population* marginal distribution, not the marginals actually seen in the sample a panel is drawn from (Section 3.2.5), so a panel satisfying population-based quotas could in principle diverge from its sample on exactly those margins, simply because the sample has drifted from the population through sampling variability or differential non-response. This is not a concern for the test behind this thesis's headline results, however: the panel-versus-sample test underlying Table 4.2's "significantly different" flag (Section 3.2.5) uses a three-way age-by-tenure-by-group interaction term whose null model already lets age and tenure vary freely by group, so only a genuine difference in the age-tenure *association* – not in either group's marginals – can trigger it. Since Panelot's quotas are marginal-only, population-based correction cannot register as a false divergence here. The exception is the standalone Cramer's V figures reported alongside this test (e.g. in Table 4.2 and Table 4.3), which are computed per table rather than as a group comparison and so remain sensitive to marginal drift.

This also explains why the survey sample, not the population, is used as the benchmark throughout this thesis: a practitioner running a real sortition process has no access to the population's *joint* age-tenure distribution to check a panel against – intersectional population data is rarely published, a gap this thesis works around rather than closes (Chapter 2) – and quotas are set from population *marginals* for the same reason. The survey sample is therefore the best source of joint information available, making the panel-versus-sample comparison used throughout Section 3.2.5 and Section 4.3 the most direct representativeness test practically achievable.

# Chapter 4

# Results

## 4.1 The problem with current visualisations

Before turning to this thesis's main results, it is worth demonstrating directly why mosaic plots – rather than the pie charts (or equivalent bar charts) that commonly accompany a sample or panel's reported composition in current sortition practice – are used throughout this chapter, rather than simply asserting it (Chapter 3).

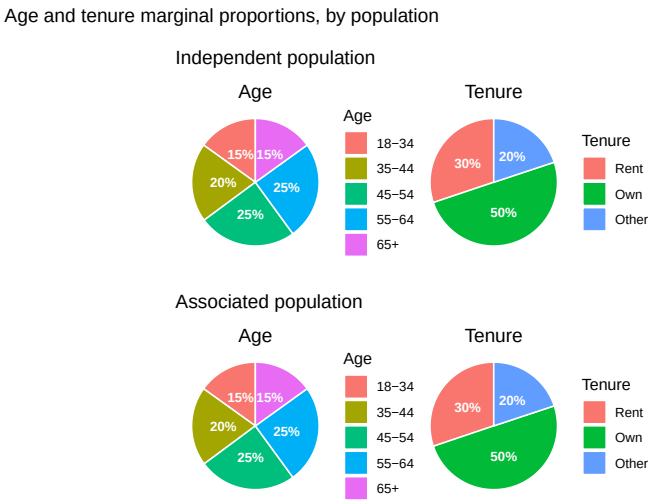



**Figure 4.1:** *Pie charts displaying age (left column) and tenure (right column) marginal proportions for the independent population (top) and associated population (bottom). The age categories range from 15%-25% and roughly half of the population owns their home. Notably, these marginal distributions are exactly the same, which hides any association.*

Figure 4.1 uses this thesis's own independent and associated populations (`sim_data2`, `sim_data2_cov`) to make the point concrete: the raking procedure used to build the associated population restores the independent population's marginals while retaining the age-renting

association introduced beforehand (Chapter 3), so the largest gap between the two populations' marginal proportions, across any single age bracket or tenure category, is only 0.02 percentage points – attributable to sampling noise alone.

Yet the two populations are not the same: the associated population carries a genuine, and statistically detectable (Table 4.1), negative association between age and renting, while the independent population has none at all. A pie chart – or a bar chart, which encodes exactly the same single-variable information in a different shape – has no visual channel available to show this: it can only display one categorical variable's marginal distribution at a time, so two variables each need their own, entirely separate chart, and nothing in either chart can reveal how the two co-occur. A practitioner shown only the pie charts in Figure 4.1 would have no way of telling the independent and associated populations apart, despite one holding a materially different – and, as Section 4.3 goes on to show, selection-relevant – relationship between age and tenure than the other. The mosaic plots used throughout the remainder of this chapter (Chapter 3) exist specifically to close this gap, representing both variables' marginal distributions and their joint, intersectional distribution within a single figure.

## 4.2 Simulation visualisations

### 4.2.1 Simulation pipeline: population, slice and respondents (equal-chance case)

Figure 4.2 and Figure 4.3 trace each population's age-tenure structure across the three pipeline stages under the equal-chance response case. Both populations skew towards upper-middle age and a higher frequency of homeownership than other categories. As the captions show, each population's age-tenure relationship (or lack of one) carries through essentially unchanged across all three stages, confirming that neither random slicing nor equal-chance response meaningfully disturbs either population's underlying structure.

### 4.2.2 Simulation pipeline: population, slice and respondents (age-associated case)

Figure 4.4 and Figure 4.5 repeat the same three-stage pipeline, but with response likelihood now rising with age rather than being equal across brackets. Unlike the equal-chance case, the age-tenure structure is no longer preserved unchanged once respondents are reached, as the captions show: an association between age and response probability – entirely independent of any association between age and tenure in the underlying population – is sufficient on its own to shift the overall age composition, and therefore the intersections present, in the final respondent sample, compounding with rather than replacing any existing age-tenure relationship in the population it is drawn from.

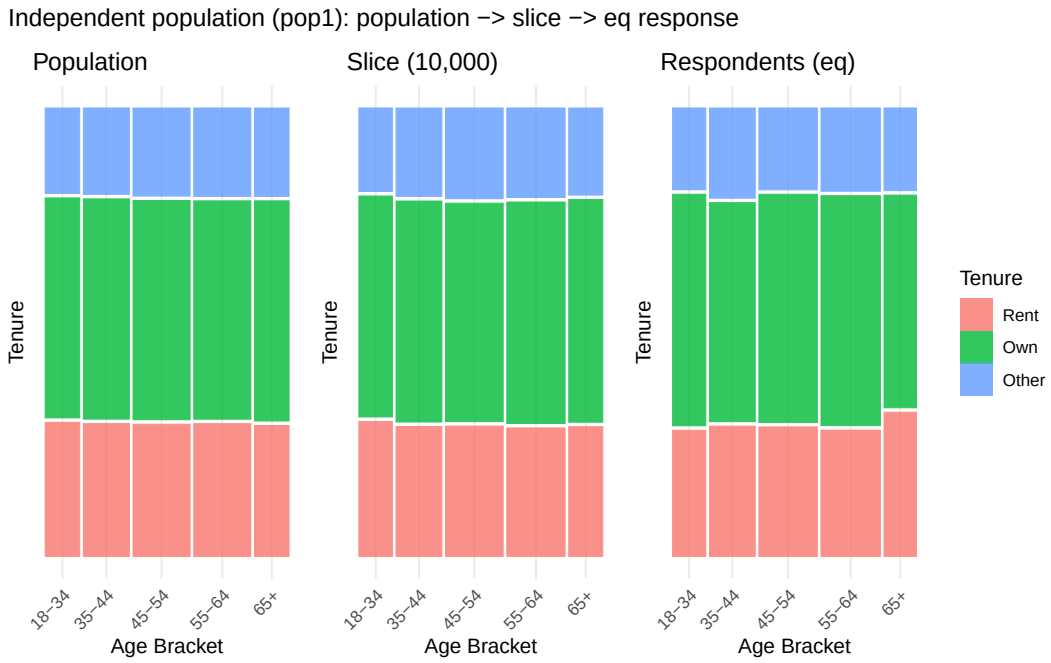



**Figure 4.2:** *Mosaic plots tracing the independent population's age-tenure structure across the population (left panel), a random 10,000-person slice (middle panel), and the slice's respondents under equal-chance response (right panel). Age brackets sit on the x-axis and tenure as fill throughout. Notice that the tile pattern barely shifts from population to respondents – there is no association between age and tenure to begin with, so neither slicing nor equal-chance response introduces one.*

### 4.2.3 Quantifying association across the pipeline

The mosaic patchworks above show the age-tenure association visually; Table 4.1 quantifies it at every stage, using a chi-squared test of independence and Cramer's V as an effect-size measure of how strong that association is.

**Table 4.1:** *Chi-squared test of independence and Cramer's V for the age-tenure association at each stage of the two-variable pipeline, for both populations and both response cases. The independent population never reaches significance at any stage (Cramer's V below 0.03 throughout), while the associated population is significant at every stage, with Cramer's V holding steady between roughly 0.125 and 0.132.*

| Population | Response case | Stage | n | Chi-sq | df | p-value | Cramer's V |
|---|---|---|---|---|---|---|---|
| Independent (pop1) | – | Population | 100000 | 5.2 | 8 | 0.735 | 0.005 |
| Independent (pop1) | – | Slice (10,000) | 10000 | 2.3 | 8 | 0.969 | 0.011 |
| Independent (pop1) | eq | Respondents | 3077 | 4.0 | 8 | 0.856 | 0.026 |
| Independent (pop1) | ass | Respondents | 3579 | 6.6 | 8 | 0.584 | 0.030 |
| Associated (pop2) | – | Population | 100000 | 3148.2 | 8 | < .001 | 0.125 |
| Associated (pop2) | – | Slice (10,000) | 10000 | 341.9 | 8 | < .001 | 0.131 |
| Associated (pop2) | eq | Respondents | 3012 | 103.4 | 8 | < .001 | 0.131 |
| Associated (pop2) | ass | Respondents | 3457 | 121.1 | 8 | < .001 | 0.132 |

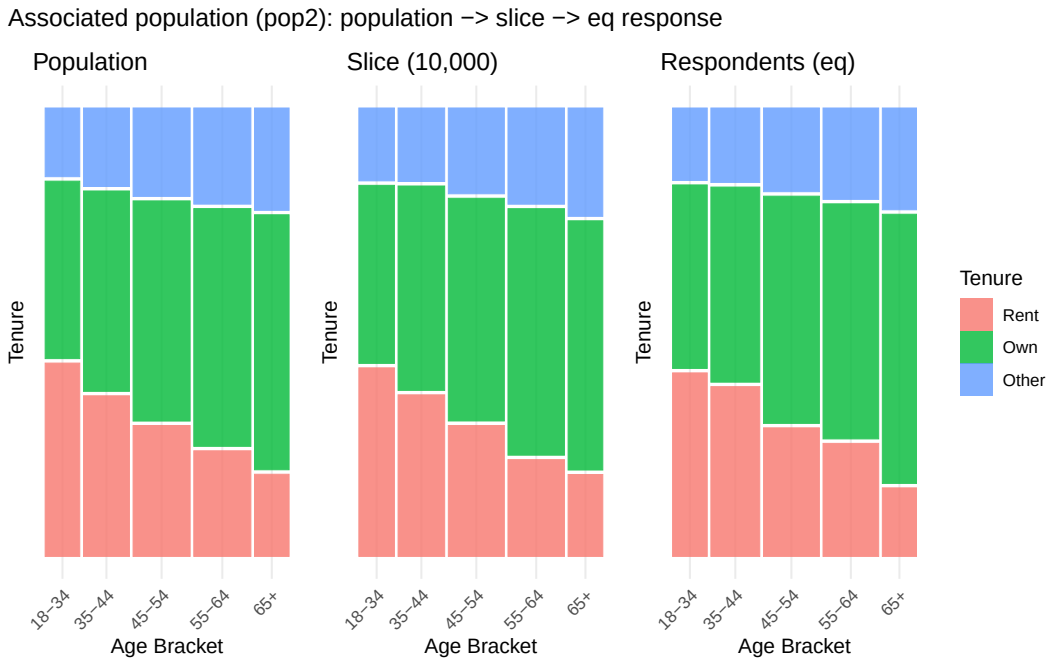



**Figure 4.3:** *Mosaic plots tracing the associated population's age-tenure structure across the population (left panel), a random 10,000-person slice (middle panel), and the slice's respondents under equal-chance response (right panel). Renting narrows steadily with age, from around 44% among 18-34 year olds to around 18% among those 65 and over. Importantly, this negative age-renting association survives essentially unchanged from population to slice to respondents, confirming equal-chance response doesn't disturb it.*

Table 4.1 confirms what Figure 4.2 through Figure 4.5 show visually: an age-tenure association that is genuinely absent in the independent population, not merely undetected, and one that is neither diluted by slicing nor meaningfully inflated by either response case in the associated population – the age-associated case's effect shows up mainly as a shift in the respondents' age *composition* (Figure 4.4, Figure 4.5), not as a change in the strength of the age-tenure relationship itself. One caveat applies to the independent population's *p*-values specifically: with population and slice sample sizes in the tens of thousands, even this chi-squared test has the power to detect very small deviations from independence, so the consistently non-significant results here reflect a genuine absence of association, not a lack of statistical power.

## 4.3 Panel-level association evaluation

### 4.3.1 Overview across lottery files

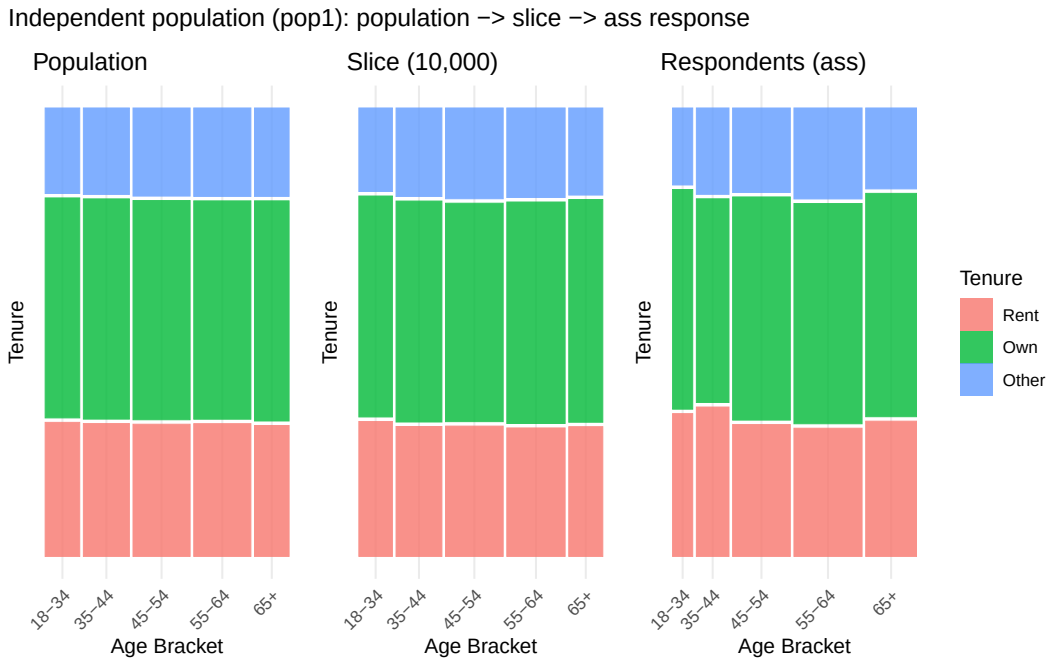



**Figure 4.4:** *Mosaic plots tracing the independent population's age-tenure structure across the population (left panel), a random 10,000-person slice (middle panel), and the slice's respondents under age-associated response (right panel), where response likelihood rises from around 20% among 18-34 year olds to over 50% among those 65 and over. Even though age and tenure carry no association in this population, the older age brackets visibly widen among respondents – differential response alone reshapes who appears, independent of any trait association in the underlying population.*

**Table 4.2:** *Panel-level association results across all four lottery files: survey-sample-level age-tenure association, and the proportion of possible panels whose association was found to be significantly different from the sample's. The proportion flagged as significantly different is saturated at or near 100% across all four files, regardless of whether the underlying population carries a genuine age-tenure association or none at all.*

| | Lottery file | Panels | Sample assoc. p | Sample Cramer's V | Prop. significantly different |
|---|---|---|---|---|---|
| 4 | lottery_ind_eq.csv | 1516 | 0.856 | 0.026 | 1.000 |
| 3 | lottery_ind_ass.csv | 3483 | 0.584 | 0.030 | 0.998 |
| 2 | lottery_cov_eq.csv | 1867 | 0.000 | 0.131 | 1.000 |
| 1 | lottery_cov_ass.csv | 2981 | 0.000 | 0.132 | 0.999 |

Table 4.2 summarises the results across all four lottery files. In the associated population, the survey sample itself retains a clearly detectable age-tenure association (Cramer's V around 0.13), consistent with the negative age-renting relationship built into the population design. In the independent population, by contrast, the sample-level association is negligible and statistically indistinguishable from zero (both $p > 0.5$), exactly as expected given no association was introduced between age and tenure at the population level.

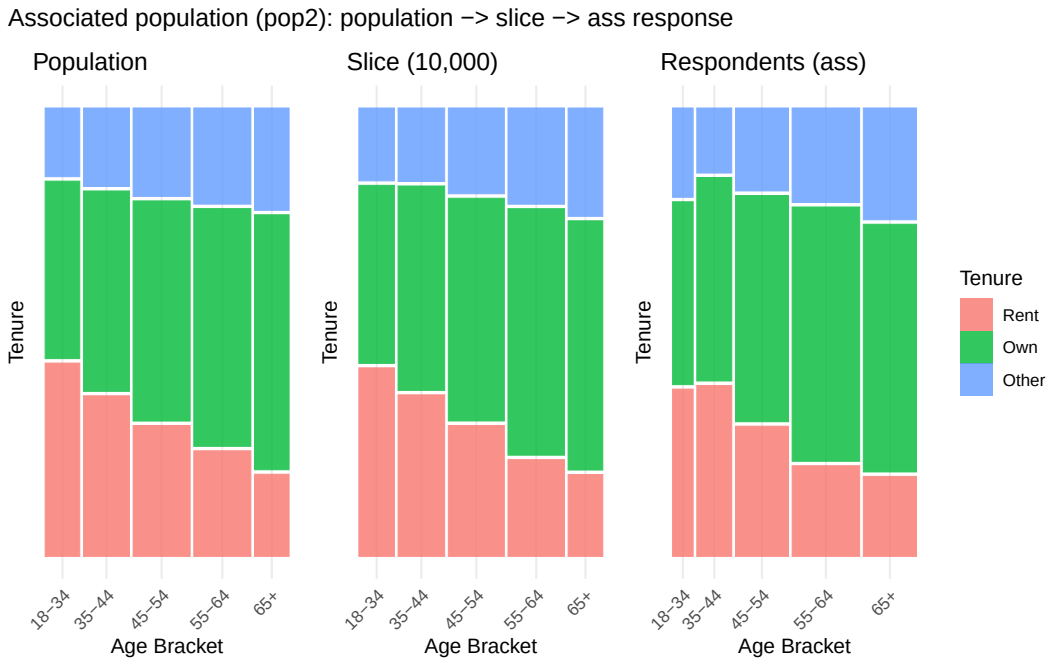



**Figure 4.5:** *Mosaic plots tracing the associated population's age-tenure structure across the population (left panel), a random 10,000-person slice (middle panel), and the slice's respondents under age-associated response (right panel). Here, the age skew from differential response compounds with the population's existing negative age-renting association rather than replacing it – respondents end up weighted even further toward the older, lower-renting end of the distribution than the slice or population alone.*

Despite this contrast in the underlying samples, the proportion of panels flagged as holding a significantly different association from their sample is strikingly similar – and strikingly high – across all four files: between 99.8% and 100%. In other words, essentially every panel Panelot could possibly return, under either population structure and either response-rate case, exhibits an age-tenure relationship detectably different from the sample it was drawn from – including in the independent population, where there is no true association for a panel to even distort.

Splitting this range apart by file shows little further structure: the two equal-chance lotteries (`lottery_cov_eq.csv`, `lottery_ind_eq.csv`) each flag 100% of their feasible panels, while the two age-associated lotteries fall only fractionally short, at 99.93% and 99.77% respectively – despite these two files containing roughly twice as many feasible panels (2981 and 3483, against 1867 and 1516), and therefore more opportunities for an occasional panel to be drawn that does not diverge from its sample. Neither population structure nor response case moves the result in a way that a difference of a few feasible panels out of several thousand cannot already explain – the distortion identified here looks like a property of the quota mechanism itself, not of any particular population or response structure it happens to be applied to.

This saturation pattern also speaks to a design concern about the panel-versus-sample test raised in

Section 3.2.5: because Panelot's quota constraints are set from each category's *population* marginal distribution rather than the survey sample's own marginals, a panel could in principle diverge from its sample simply because the sample's composition has drifted from the population's, rather than because of anything specific to the selection mechanism. Under equal-chance response, however, the survey sample's age marginal already matches the population's closely (Figure 4.2, Figure 4.3), leaving barely any population-versus-sample gap for population-based quotas to correct – yet the equal-chance files above are flagged as different *at least as often* as the age-associated ones, where the sample's marginal has genuinely drifted from the population's (Figure 4.4, Figure 4.5). If population correction were driving the divergence, equal-chance panels should have been flagged less often, not equally or more. Together with the test's own design – which already allows panel and sample to differ freely on each marginal without that alone counting as significant (Section 3.2.5) – this confirms that the near-total divergence documented here reflects the quota mechanism leaving the joint age-tenure distribution unconstrained, rather than any population-versus-sample drift.

### 4.3.2 Comparison against the random-sampling benchmark

**Table 4.3:** *Proportion of panels (or random samples) whose age-tenure association was found to be significantly different from the survey sample's, and the median within-group Cramer's V, for actual sortition panels versus 500 simple random samples of equal size, for all four lottery files. Sortition panels are flagged as different an order of magnitude more often than equivalently sized random samples, with roughly three times the median Cramer's V.*

| Lottery file | Panels | Prop. sig. diff. (panels) | Median V (panels) | Prop. sig. diff. (SRS) | Median V (SRS) |
|---|---|---|---|---|---|
| cov_eq | 1867 | 1.000 | 0.878 | 0.084 | 0.301 |
| cov_ass | 2981 | 0.999 | 0.878 | 0.082 | 0.300 |
| ind_eq | 1516 | 1.000 | 0.876 | 0.094 | 0.274 |
| ind_ass | 3483 | 0.998 | 0.873 | 0.084 | 0.273 |

Table 4.3 repeats the comparison introduced in Section 3.2.5 across all four lottery files, rather than the single associated/equal-chance case alone. Actual sortition panels are flagged as significantly different from their sample 99.8-100% of the time, against only 8.2-9.4% for equivalently sized random samples – modestly above the nominal 5% false-positive rate expected under the null (plausibly the homogeneity test's asymptotic approximation being somewhat liberal at this cell count, rather than genuine divergence from random sampling). Median within-group Cramer's V tells the same story: panels sit around 0.87-0.88, against 0.27-0.3 seen from random draws, in every file regardless of population or response case. Because the benchmark result holds uniformly, rather than being specific to the associated/equal-chance file used as the illustrative case above, the explanation cannot be population- or response-case-specific either – reinforcing that it is Panelot's quota mechanism, not

sampling variability at panel size or any one simulated condition, driving the divergence documented in Table 4.2.

### 4.3.3 Illustrative example

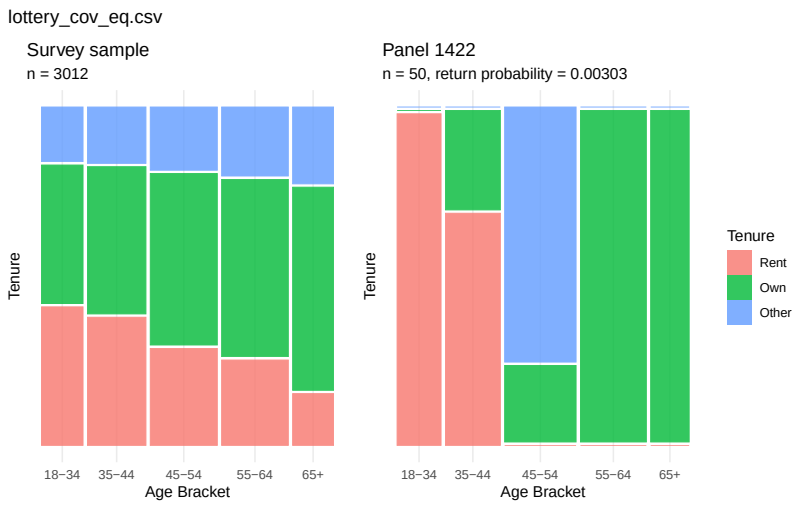



**Figure 4.6:** *Age-tenure composition of the associated population's survey sample (left) alongside its most likely panel under lottery_cov_eq.csv (right). Although the panel satisfies the same marginal age and tenure quotas as the sample, its age-tenure combinations are considerably more blocky and uneven – several age brackets end up filled by entirely one tenure category, exaggerating the sample's already-present age-renting association.*

Figure 4.6 makes explicit what the aggregate figures above describe in summary: the smooth, gradually strengthening sample-level relationship between age and renting gives way to a considerably more extreme, blocky panel-level pattern, despite the panel satisfying the same marginal quotas as the sample it was ostensibly drawn to represent.

Figure 4.7 shows the same comparison for the independent population. The panel is, if anything, more extreme than the associated-population example: the within-panel test gives a Cramer's V of 0.9, versus 0.03 for the sample it was drawn from. Because this panel's distorted joint structure cannot be attributed to a genuine association it is "exaggerating" – there is none to exaggerate in the independent population – it most directly demonstrates that the within-panel association test is picking up an artefact of the selection mechanism, not a real signal faithfully (if imperfectly) carried through from the sample.

### 4.3.4 Summary

Across all four simulated conditions, satisfying Panelot's marginal quotas on age and tenure did not translate into a panel whose joint age-tenure structure resembled its survey sample's: essentially every feasible panel was flagged as significantly different from the sample, at a rate far beyond what panel size alone could explain, and this held regardless of whether the underlying population carried

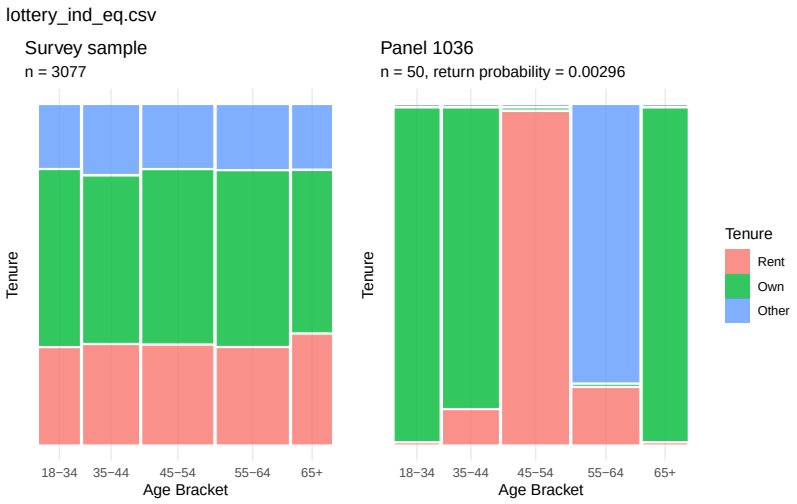



**Figure 4.7:** *Age-tenure composition of the independent population's survey sample (left) alongside its most likely panel under lottery_ind_eq.csv (right). Even though age and tenure carry no association in the sample – its fill proportions sit close to flat across every age bracket – the panel still ends up blocky and uneven, with most age brackets dominated by a single tenure category, showing the distortion is an artefact of selection itself rather than an exaggeration of a real trend.*

a genuine age-tenure association or response was skewed by age. Marginal quota satisfaction – the only form of representativeness that a current sortition tool like Panelot reports to a practitioner – is therefore not a reliable proxy for intersectional representativeness in this simulation, and the gap between the two looks like a structural consequence of leaving the joint distribution unconstrained, rather than an artefact of this thesis's particular panel size or association strengths. This divergence is not attributable to panels merely correcting toward population marginals where the sample itself has drifted – a possibility already ruled out above, both by the panel-versus-sample test's design and by the data (Section 4.3; Chapter 3). Chapter 5 introduces Intersection Inspector, a tool built directly in response to this marginal-versus-intersectional gap, which displays a panel's intersectional association alongside its marginal quota satisfaction so that this kind of distortion is visible to a practitioner before a panel convenes, rather than only to a reader of this thesis after the fact.

# Chapter 5

# Introducing Intersection Inspector

Having flagged the issues with current sortition methods and process, the second gap this thesis set out to address was the absence of a tool for practitioners to assess the viability and intersectionality of panels before deliberation begins. To that end, a Shiny app was developed. Even in current practice, the LexiMin lottery and panel selection process takes two steps: uploading categories, trait quotas and a respondent pool with the desired panel size, then taking the resulting lottery to a separate interactive page for random number generation to select a specific panel. This thesis's app replaces that second step, with added functionality that smooths a practitioner's workflow and gives them more confidence in a panel's representativeness at a controllably granular level of intersectionality – enabling more informed decisions about quota adjustments and representative validity, and reducing the likelihood of unexpectedly poor intersectional spread when panels convene.

## 5.1 Context: prior steps in the sortition pipeline

Intersection Inspector is not intended to replace the sortition process as a whole, only its final step. By the time a practitioner opens the app, they are assumed to have already carried out the preceding stages of a sortition: recruiting and surveying a respondent pool, specifying the category variables and trait quotas the panel should satisfy (e.g. minimum and maximum counts per age bracket, per tenure category), and running a selection algorithm over that pool to discover every panel of the desired size that satisfies those quotas. In current practice this selection step is performed by LexiMin, through the Panelot app hosted at Panelot.org, which outputs a lottery file enumerating each feasible panel together with its probability of being returned.

The app takes this lottery file and the underlying respondent pool as given, and does not re-run or second-guess the algorithm that produced them. Its usefulness is therefore not tied to LexiMin

specifically: any sortition algorithm or process producing an equivalent output – a set of feasible panels (or a single panel) drawn from a known respondent pool, satisfying some quota specification – could be inspected through the app the same way. The quota-based selection step and the intersectional evaluation step addressed here are conceptually separate, and the latter is agnostic to which algorithm performed the former.

## 5.2 Intended workflow

The app page consists of three parts: 1. a sidebar comprised of interactable buttons, toggles and sliders, 2. a window that displays mosaic plots comparing the 2-way intersections and associations present between the sample and any selected panel for any two selected category variables, and 3. a window that displays the relative proportions of every trait present in the sample and any selected panel, across all categories present within the sample's dataset. Figure 5.1 shows all three parts together, using the associated population's equal-chance lottery (`lottery_cov_eq.csv`) – the same example used for Figure 4.6 in Chapter 4 – as the uploaded sample and lottery file. To make use of this, a practitioner is expected to follow the below procedure.

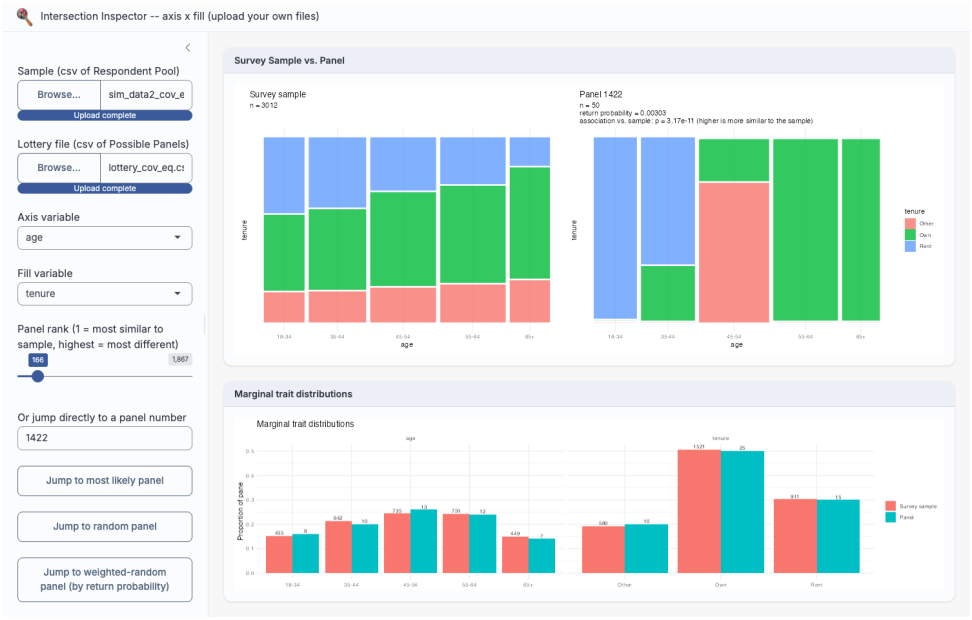



**Figure 5.1:** *Intersection Inspector's main view: the sidebar (left) for file upload, variable selection and panel navigation; the "Survey Sample vs. Panel" mosaic comparison (top right); and the "Marginal trait distributions" bar chart (bottom right). Shown here with the same associated-population, equal-chance example (`lottery_cov_eq.csv`) used in Figure 4.6.*

### 5.2.1 Uploading files

Intersection Inspector takes two files as input: the sample file the lottery was drawn from (the respondent pool that was invited and agreed to participate) and a lottery file (Panelot's `lottery.csv`,

enumerating every panel the sortition algorithm found feasible, each with its member IDs and return probability). From these two files alone, the app reconstructs every selected panel's composition on each category present in the sample and makes its intersectional structure directly inspectable, without the practitioner writing any code or leaving the browser.

### 5.2.2 Required file structure

Because the app is designed to work with any sortition process producing an equivalent output, not just this thesis's own simulation (Chapter 5), it does not hard-code column names or category domains. Instead, it expects each uploaded file to follow a general structure – the same structure Panelot itself produces, and the one this thesis's own simulated data (Chapter 3) is written out in.

The **sample file** (the respondent pool, or survey sample, a lottery was drawn from) should hold one row per respondent, with:

- an optional ID column, detected case-insensitively by header name (`pool_member_identifier`, `id`, `member_id`, `respondent_id`, or `identifier`); if none of these is found, respondents are instead identified by row number, and lottery member IDs are matched against row position;
- two or more further columns, each treated as a single categorical trait (e.g. age bracket, tenure, gender, region). Values in these columns are coerced to character and sorted alphabetically to form each trait's factor levels, since the app has no way to infer a meaningful domain-specific order (e.g. "18-34" before "35-44") for an arbitrary uploaded variable – a limitation with consequences for ordinal variables discussed further in Section 5.3.

For example, this thesis's own `sim_data2_cov_eq.csv` begins:

```
"pool_member_identifier","age","tenure"
33,"18-34","Rent"
67,"18-34","Rent"
```

The **lottery file** (Panelot's `lottery.csv`, or equivalent) should hold one row per *possible* panel, with:

- a "return probability of panel" column (if applicable), giving the probability panel $j$ would be returned were the lottery run – optional for the mosaic plots and association tests, but required for the weighted-random panel button;
- one "panel member 1", "panel member 2", ... column per selected panel member, each holding that member's ID, matched against the sample file's ID column (or row number, if no ID column was detected).

Correspondingly, `lottery_cov_eq.csv` begins:

```
return probability of panel,panel member 1,panel member 2,panel member 3,...
0.000890,4957,4961,4964,4966,5017,...
0.001015,6132,6166,6181,6239,6284,...
```

Every trait column in the sample file, not just the two currently selected for the mosaic plot, appears in the marginal bar chart described below (Figure 5.3) – so a practitioner can switch which two variables the mosaic plots compare without re-uploading anything.

### 5.2.3 Comparing intersections and marginal proportions

Once the files are uploaded, the practitioner selects any two category variables of interest (e.g. age and tenure) and the app renders a pair of mosaic plots side by side: one for the currently selected panel, one for the sample it was drawn from. These mosaic plots visualise the two-way intersection and association between the chosen variables, so that a practitioner can see at a glance whether the panel's joint distribution of traits resembles the sample's "shape", or whether it is skewed in a way that marginal quota counts alone would not reveal. Figure 5.2 shows this comparison for the same most-likely panel examined manually in Figure 4.6 – the app reproduces that same blocky, uneven age-tenure structure directly, alongside the panel's association test p-value, without requiring any code. Beneath the mosaic plots, a second window displays bar charts of every trait's marginal proportion in the panel against the sample, across every category in the dataset, covering representativeness at the single-variable level as well.




**Figure 5.2:** *The "Survey Sample vs. Panel" mosaic comparison, showing the most likely panel under* `lottery_cov_eq.csv` *against its survey sample – the same panel and sample compared manually in Figure 4.6.*

Figure 5.3 shows the accompanying "Marginal trait distributions" view for the same panel and sample: every age bracket and tenure category side by side, with counts labelled directly on each bar. Where the mosaic plots above expose the panel's distorted *joint* structure, this view confirms the panel's single-variable marginals stay close to the sample's throughout – precisely the gap between marginal and intersectional representativeness this thesis is concerned with (Section 4.1).

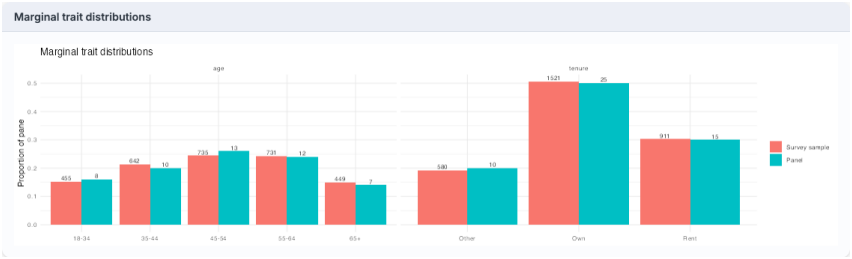


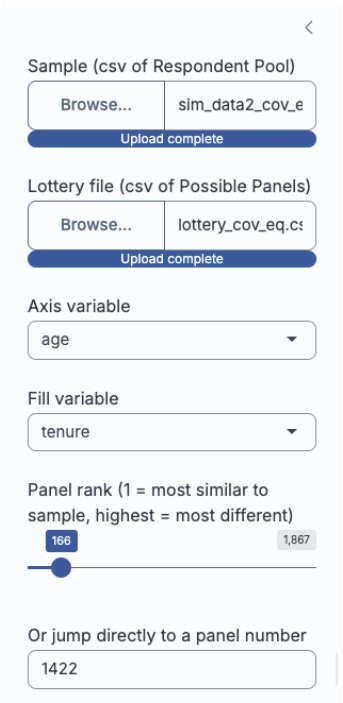

**Figure 5.3:** *The "Marginal trait distributions" view, showing every age and tenure category's proportion in the survey sample against the same most-likely panel under* `lottery_cov_eq.csv` *used in Figure 5.2.*

### 5.2.4 Testing and browsing panels

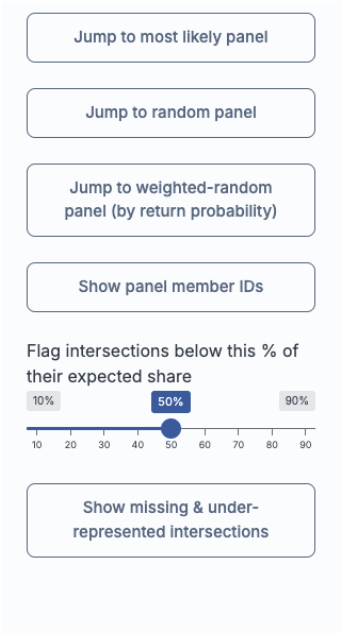
Beyond visual inspection, the app formally tests association preservation for each panel using the tests described in Section 3.2.5, summarising each as a p-value and significance interpretation ("level of difference" from the sample), alongside the panel's size and its probability of selection by the lottery. Panels can be browsed in several ways: sliding across all discovered panels ordered from most to least similar in association to the sample, jumping to a uniformly random panel (for arbitrary evaluation of the algorithm's performance), jumping to a probability-weighted random panel (sidestepping the need for a separate external random number generator once a panel is to be finalised), or entering a specific panel ID directly. A further view lets the practitioner check, at an adjustable representation threshold, which intersections are missing or under-represented in the panel relative to the sample. Once a panel is chosen, its member IDs can be inspected and copied to the clipboard to continue the existing workflow (e.g. contacting selected members). Figure 5.4 shows the full set of sidebar controls behind this workflow, from file upload and variable selection down to panel navigation, member ID lookup, and the under-representation threshold slider.

### 5.2.5 Replacing the Panelot.org selection step

In current practice, after a lottery file is generated, a practitioner takes it to a second, separate interactive page – on Panelot.org – purely to generate a random number and select one specific panel from those enumerated, then manually looks up the member IDs from the output. That step reveals nothing about *which* panel is being selected or how representative it is beyond the quota constraints already satisfied; it is, in effect, a black box that turns a lottery file into a single panel. Intersection Inspector is designed as a drop-in replacement: it accepts the same sample and lottery files, supports the same random (and probability-weighted random) selection as a final action, and returns the same member list a practitioner needs – while additionally providing the mosaic plots, marginal bar charts, association p-values, and under-representation checks along the way.




**(a)** *File upload, axis/fill variable selection, panel-rank slider and direct panel-ID entry.*




**(b)** *Random and weighted-random jump buttons, panel member ID lookup, and the adjustable under-representation threshold.*

**Figure 5.4:** *The sidebar's full set of controls, split top-to-bottom into two columns for readability.*

### 5.2.6 Benefits for new practitioners

This is particularly intended to benefit practitioners who are new to sortition or without a statistics background. Rather than trusting that satisfied quota constraints automatically mean a representative panel, the practitioner is given an interpretable visual and numerical basis for that trust – or for flagging concern – before a panel is convened. Because panels can be scrolled through and compared on their association preservation, the app can also be used diagnostically: if every available panel shows a similarly poor association to the sample, that points to the quota constraints themselves needing revision, rather than to any single unlucky draw, and may prompt the practitioner to consider whether additional surveying is needed for a more representative sample. Where a blindspot is genuinely infeasible for sortition to resolve on its own – an intersection too thin in the respondent pool for any quota-constrained panel to capture, say – the tool at least makes that limit visible, so the practitioner can weigh up whether a separate process (targeted outreach, or a specialist contribution alongside the panel) is needed to fill the gap. In this way, the app turns a previously invisible step of the sortition pipeline into a transparent, inspectable one, directly in the hands of the people already running it.

## 5.3 Current limitations

The mosaic plots and association tests operate on exactly two category variables at a time, chosen from the sidebar's axis and fill dropdowns (Figure 5.4); a dataset with more trait columns – age, tenure, gender, and region, say – cannot have its full joint structure inspected in a single view, so a practitioner steps through pairs of interest instead. This is a genuine constraint relative to the full joint distribution a panel's representativeness ultimately depends on, not a design oversight: a fully faceted view spanning four or more variables would trade interpretability for density, and a practitioner without a statistics background is least likely to benefit from that trade. Restricting the view to two variables at a time instead gives deliberate, granular control over the "traits that matter" for a given engagement, rather than every combinatorial intersection a dataset happens to contain, most of which may be irrelevant to the community context at hand – at the cost of checking each pair in turn rather than seeing the full picture at once, a trade-off future versions could revisit by letting a practitioner flag priority combinations to surface automatically.

A further limitation stems from how uploaded trait columns are turned into factor levels: as noted above (Section 5.2.2), values are coerced to character and sorted alphabetically, since the app has no way to infer a meaningful domain-specific order for an arbitrary uploaded variable. This is immaterial for a nominal variable (gender, region, tenure), but for an ordinal variable it can produce a level order that does not match the variable's natural progression – an income bracket column with levels "Under $20k", "$20k-$50k", "$50k-$100k", "Over $100k" would not sort correctly alphabetically, nor would an agreement-scale variable running "Strongly disagree" through "Strongly agree".

This does not affect the app's core purpose: the mosaic plots and association tests operate correctly regardless of level order, since both the panel's and sample's blocks are drawn from the same (mis)ordering and remain directly comparable. What it does affect is a mosaic plot's readability by eye – an ordinal variable with shuffled levels can look "patchier" or less monotonic than the underlying relationship warrants, since blocks that would otherwise sit adjacent and show a smooth gradient are instead scattered across the axis. A practitioner inspecting such a plot should check whether either selected variable is ordinal and read it with that re-ordering in mind, rather than inferring a lack of structure from a sorting artefact. Addressing this within the app – letting a practitioner manually specify a level order at upload time – would require per-column metadata the app's intentionally format-agnostic file structure does not currently collect, and so is left for future refinement.

Taken together, these limitations do not undercut the app's core contribution: Intersection Inspector still closes the central gap this thesis has identified, putting a panel's joint structure in front of a practitioner for the first time rather than leaving it unexamined behind satisfied marginal quotas.

# Chapter 6

# Conclusion

This thesis set out from a deceptively simple question: when a sortition algorithm satisfies every quota it is given, does the panel it returns actually represent the population it was drawn from? The literature reviewed in Chapter 2 showed that decades of algorithmic development – from LEGACY through MaxiMin and MiniMax to LexiMin and Goldilocks – have answered an adjacent but different question extensively: how to make individual selection probabilities as fair and as robust to manipulation as possible. What none of these methods do is stratify, constrain, or even check their output against the *combinations* of traits that make up a population's intersectional structure. The implicit assumption running through the field has been that marginal fairness, achieved at the level of each demographic category independently, is sufficient to carry intersectional representativeness along with it for free.

Chapter 3 and Chapter 4 tested that assumption directly, and found it does not hold. Across every simulated condition – independent or associated populations, equal-chance or age-skewed response – effectively every feasible panel Panelot's LexiMin implementation could return showed an age-tenure relationship significantly different from its own survey sample, at a rate roughly an order of magnitude above what simple random sampling of the same size produces. Strikingly, this held even in the independent population, where there was no true association for a panel to represent in the first place: the distortion is a property of the quota mechanism itself, not of any one population structure or non-response pattern. Panels built to satisfy marginal quotas on age and on tenure separately were, time and again, blocky and uneven in their joint composition – a pattern invisible to the pie charts and bar charts that typically accompany reported panel composition in practice (Section 4.1), and invisible, more fundamentally, to any quota system that only ever looks at one trait at a time.

Chapter 5 responded to that finding with Intersection Inspector: a tool aimed at closing the gap between what current selection algorithms guarantee and what practitioners actually need to know

before convening a panel. By placing mosaic plot comparisons, marginal bar charts, and formal association tests directly in a practitioner's hands – alongside panel browsing, random and weighted-random selection, and an under-representation threshold – the app turns the previously invisible final step of the sortition pipeline into something inspectable, without requiring a statistics background to use.

## 6.1 Framework and recommendations

Taken together, these chapters suggest a straightforward recommendation for practitioners and tool-builders alike: marginal quota satisfaction should be treated as necessary, but never sufficient, evidence of a representative panel. A panel that hits every category count it was asked to hit can still systematically misrepresent how those categories combine in the community it is meant to reflect, and nothing in the quota-setting or selection process as currently practised will achieve that on its own.

In practical terms, this points toward a layered framework for panel evaluation. First, quota-based selection (LexiMin, or in future, Goldilocks) remains the right tool for ensuring fair and robust selection probabilities at the marginal level – this thesis does not dispute the value of that work, only its sufficiency. Second, an intersectional check, of the kind Intersection Inspector performs, should sit between panel generation and panel convening, so that a practitioner can see whether a panel's joint trait structure plausibly reflects its sample before members are contacted, rather than discovering a skew only once deliberation is already underway. Third, where a specific intersectional combination is known to matter for an engagement's context – a particular minority subgroup, say – this thesis's results suggest that relying on marginal quotas to deliver it incidentally is optimistic at best; deliberate joint-quota constraints, or supplementary recruitment targeted at that intersection, are more defensible routes than trusting the selection mechanism to produce it as a byproduct.

## 6.2 Limitations

Several limitations bound how far these findings should be read. The simulation used two categorical variables – age and tenure – chosen for their resemblance to variables genuinely used in sortition practice, but real panels are typically built against three, five, or more demographic categories simultaneously; the discrepancy documented here is expected to compound, not relax, as the intersectional space grows while panel size stays fixed. However, that expectation was not itself tested at higher dimensionality (Chapter 3). Likewise, response rates were deliberately elevated well above what civic lotteries typically see in the field, to keep respondent cells populated enough to isolate the algorithm's

own behaviour; this makes the reported distortion a conservative, best-case estimate rather than an upper bound, and the association strengths introduced into the simulated populations were fixed at values chosen for visibility rather than drawn from, or validated against, any specific real-world dataset (Chapter 3).

Intersection Inspector itself carries its own, openly acknowledged constraints (Section 5.3). It inspects two variables' joint structure at a time, rather than the full multivariate intersection a panel's representativeness ultimately rests on. It also sorts uploaded trait levels alphabetically, which can misorder ordinal variables and make an association harder to read by eye – though the underlying statistical tests remain valid regardless of level order. Both are deliberate trade-offs in favour of interpretability for non-specialist users, but they mean the app diagnoses the problem this thesis identifies more thoroughly than it solves it outright.

## 6.3 Future research

A natural next step is extending the simulation framework beyond two variables, to establish whether, and how severely, the intersectional distortion observed here scales with additional demographic categories – and whether it does so uniformly or concentrates in particular combinations with many trait levels. The Goldilocks algorithm, not yet publicly available at the time of writing, is also worth revisiting once released: it addresses a different problem (robustness to manipulation) to the one this thesis raises, but it remains an open question whether its joint handling of selection-probability ceilings and floors incidentally improves, worsens, or leaves unchanged the intersectional preservation documented here. More broadly, this thesis's central finding motivates a more direct line of work: selection methods that explicitly incorporate joint, intersectional constraints alongside marginal quotas, rather than treating intersectionality purely as something to check for after the fact. One emerging example is Diversimax (Matar 2025), which uses mixed integer programming to select panels that explicitly minimise imbalance across every intersectional combination of demographic traits, rather than optimising selection probabilities at the marginal level alone; it is not evaluated in detail here, since it has yet to see the kind of practitioner adoption LexiMin has. However, it is also worth treating cautiously rather than as a straightforward replacement: optimising for intersectional balance alone, without also bounding how unequal individual selection probabilities can become, risks reopening the same extreme-probability and manipulation vulnerabilities that motivated MaxiMin, MiniMax, LexiMin and Goldilocks in the first place – a panel could end up intersectionally balanced only because some individuals were given a far higher selection chance than others to fill sparse cells. Reconciling intersectional balance with that fairness guarantee looks like the more productive target

than optimising for either alone.

On the tooling side, a practical next step is packaging this thesis's LexiMin implementation as a proper R package and integrating panel selection directly into Intersection Inspector, rather than requiring a practitioner to generate a lottery file via the separate Panelot.org interface before uploading it to the app. Collapsing today's two-tool workflow into a single environment – covering both panel generation and intersectional evaluation – would remove a point of friction for practitioners, and would make it straightforward to extend the same integration to Goldilocks or Diversimax as those become more broadly available.

## 6.4 Closing remarks

It is worth being plain about what this thesis is, and is not, arguing. The fairness and robustness advances surveyed in Chapter 2 are genuine achievements, and nothing here suggests abandoning them. LexiMin's maximisation of selection probabilities, and Goldilocks' guard against manipulation, solve real problems that mattered enough to motivate a decade of algorithmic development. But a selection *mechanism* being fair is a different property from the *panel* it produces being representative, and this thesis has tried to show, explicitly, that the gap between the two is not merely theoretical. Optimising who gets a chance to be selected is not the same question as asking who, collectively, ends up involved – and whether the particular mix of people assembled actually reflects the community whose voice the process claims to amplify.

Representation, in other words, is not just a matter of getting the proportions right on each axis taken separately; it is a matter of getting the *people* right, in all the ways their identities and circumstances combine. A panel can be perfectly fair by the numbers and still quietly leave an entire slice of a community's lived experience unheard, simply because no one checked whether the categories were ever meant to be looked at together. If sortition's promise is a seat for everyone, a fair lottery is not the end of the question – it is only the beginning. The real test is who actually ends up on the panel, with their thoughts and feelings heard. So before the next panel is called fair and finished, it is worth pausing to ask: am I in the room?

# Bibliography


Baharav, C & B Flanigan (2024). Fair, Manipulation-Robust, and Transparent Sortition. *Proceedings of the 25th ACM Conference on Economics and Computation*, 756–775. (Visited on 04/15/2026).

Chang, W, J Cheng, J Allaire, C Sievert, B Schloerke, G Aden-Buie, Y Xie, J Allen, J McPherson, A Dipert & B Borges (2026). *shiny: Web Application Framework for R*. R package version 1.14.0. https://shiny.posit.co/.

Flanigan, B, P Gölz, A Gupta, B Hennig & AD Procaccia (2021). Fair Algorithms for Selecting Citizens' Assemblies. *Nature* **596**, 548–552.

Flanigan, B, P Gölz, A Gupta, AD Procaccia & G Rusak (2020). *Panelot*. Online implementation of the LexiMin algorithm for sortition panel selection. https://panelot.org.

Flanigan, B, G Kehne & AD Procaccia (2021). Fair Sortition Made Transparent. In: *Advances in Neural Information Processing Systems*. Vol. 34, pp.25720–25731. https://procaccia.info/wp-content/uploads/2021/05/transparent.pdf.

Friendly, M, G Klorfine, H Jeppson, H Hofmann & D Cook (2026). *ggmosaic2: Mosaic Plots in the 'ggplot2' Framework, Extended*. R package version 0.5.1. https://friendly.github.io/ggmosaic2/.

Giuliani, A (2025). Sortition in Politics: From History to Contemporary Democracy. *Parliaments, Estates and Representation* **45**(3), 317–335.

Hofmann, H (2003). Constructing and Reading Mosaicplots. *Computational Statistics & Data Analysis* **43**(4), 565–580.

Jeppson, H, H Hofmann & D Cook (2026). *ggmosaic: Mosaic Plots in the ggplot2 Framework*. R package version 0.4.0. https://haleyjeppson.github.io/ggmosaic/.

Landemore, H (2026). Opinion | No Shy Person Left Behind. *The New York Times*. (Visited on 04/20/2026).

Matar, O (2025). *Diversimax: Maximizing Intersectional Diversity in Sortition-Based Citizen Assemblies*. Tech. rep. Democracy 3.0. https://orenmatar.github.io/diversimax_algorithm/.

Paris Creates a Permanent Citizens' Council (2021). *Sortition Foundation*. (Visited on 04/14/2026).

Pedersen, TL (2025). *patchwork: The Composer of Plots*. R package version 1.3.2. https://CRAN.R-project.org/package=patchwork.

R Core Team (2026). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/.

Redman, K & L Carson (n.d.). How to Recruit Participants for Citizens' Assemblies ().

Sortition Foundation (2024). *GroupSelect App*. https://github.com/sortitionfoundation/groupselect-app.

Ushey, K & H Wickham (2025). *renv: Project Environments*. R package version 1.1.5. https://rstudio.github.io/renv/.

Walsh, CD & JA Elkink (2021). The Dissatisfied and the Engaged: Citizen Support for Citizens' Assemblies and Their Willingness to Participate. *Irish Political Studies* **36**(4), 647–666. (Visited on 04/15/2026).

Ward, I (2008). An Experiment in Political Communication: The British Columbia Citizens' Assembly on Electoral Reform. *Australian Journal of Political Science* **43**(2), 301–315. (Visited on 04/15/2026).

Wickham, H (2023). *tidyverse: Easily Install and Load the Tidyverse*. R package version 2.0.0. https://tidyverse.tidyverse.org.

Wickham, H, W Chang, L Henry, TL Pedersen, K Takahashi, C Wilke, K Woo, H Yutani, D Dunnington & T van den Brand (2026a). *ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics*. R package version 4.0.3.9000. https://ggplot2.tidyverse.org.

Wickham, H, R François, L Henry, K Müller & D Vaughan (2026b). *dplyr: A Grammar of Data Manipulation*. R package version 1.2.1. https://dplyr.tidyverse.org.

Wickham, H, D Vaughan & M Girlich (2025). *tidyr: Tidy Messy Data*. R package version 1.3.2. https://tidyr.tidyverse.org.

Xie, Y (2025). *knitr: A General-Purpose Package for Dynamic Report Generation in R*. R package version 1.51. https://yihui.org/knitr/.

Zhu, H (2026). *kableExtra: Construct Complex Table with kable and Pipe Syntax*. R package version 1.4.1. https://haozhu233.github.io/kableExtra/.

# Appendix A

# Iterative proportional fitting (raking) derivation

This appendix gives the full iterative update rule underlying the raking step summarised in Chapter 3, used to restore the age and tenure marginal targets $\mathbf{p}_A$ and $\mathbf{p}_T$ after the associated population's exponential seeding adjustment (Equation 3.4) disturbs them.

Initialising $P^{(0)}_{\text{rake}} = \tilde{P}$, each raking iteration $r = 0, 1, 2, \ldots$ performs two rescaling passes,

$$P^{(r+1/2)}(a,t) \;=\; P^{(r)}_{\text{rake}}(a,t) \cdot \frac{p_{A_a}}{\sum_{t'} P^{(r)}_{\text{rake}}(a,t')}, \qquad P^{(r+1)}_{\text{rake}}(a,t) \;=\; P^{(r+1/2)}(a,t) \cdot \frac{p_{T_t}}{\sum_{a'} P^{(r+1/2)}(a',t)}, \tag{A.1}$$

first rescaling every row to sum to $p_{A_a}$, then every column to sum to $p_{T_t}$. Repeating this pair of passes for $R = 100$ iterations (well past convergence for a $5 \times 3$ table) yields the raked array $P^{\star}$, satisfying

$$\sum_t P^{\star}(a,t) = p_{A_a} \;\text{ for all } a, \qquad \sum_a P^{\star}(a,t) = p_{T_t} \;\text{ for all } t, \tag{A.2}$$

to numerical tolerance, while retaining the negative age-renting association introduced by the exponential seeding step – raking restores the specified marginals while preserving the association the adjustment introduced.